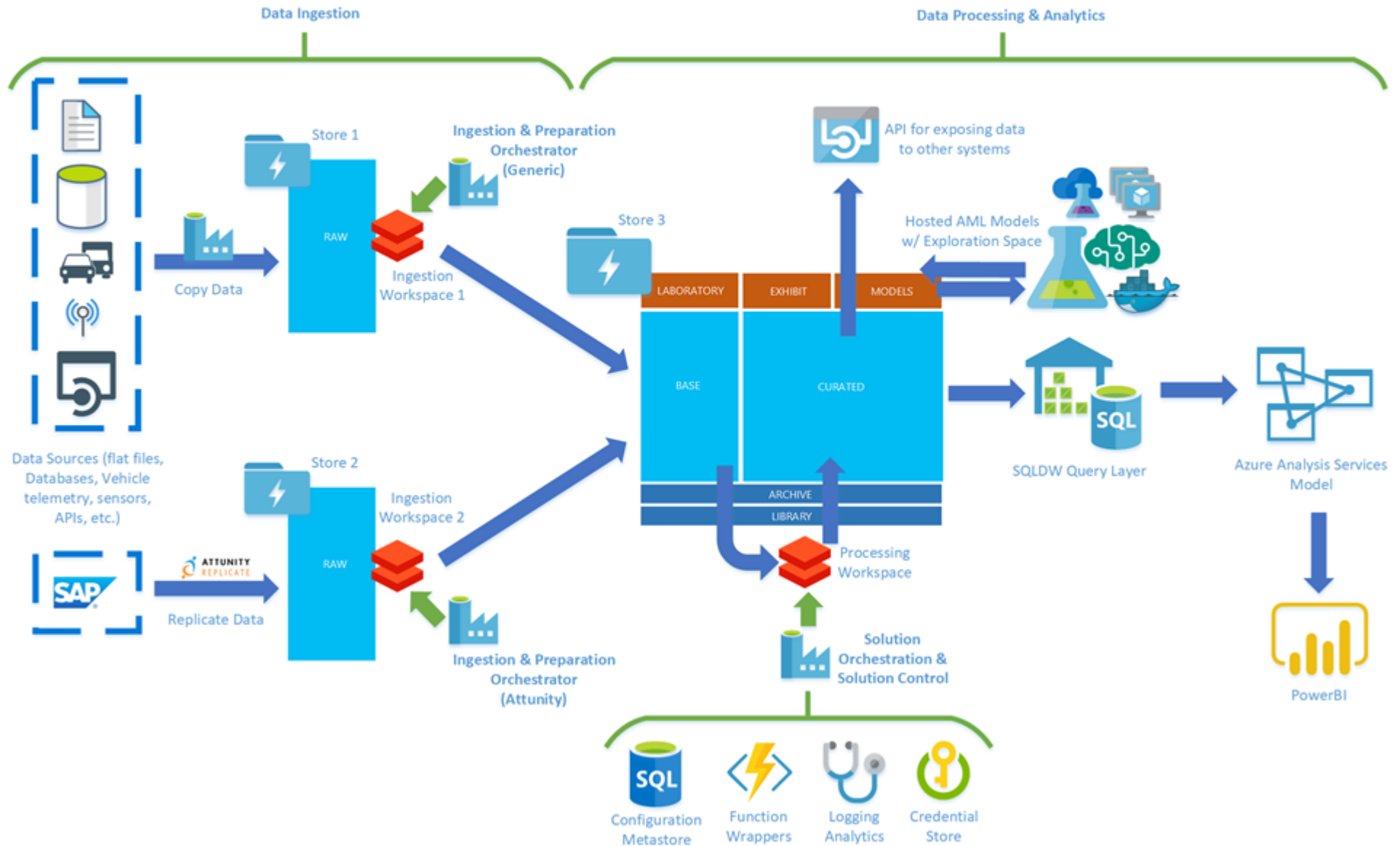


Last updated by | Markham Neil | Jul 18, 2019 at 8:04 AM GMT+1

- Introduction
- Planned Orchestration
- Source to Raw
- Raw to Base
- Base to Curated
- Curated to Semantic

This page illustrated the start to end orchestration for Telematics and Data Science. Refer the [Design Overview and Approach](#) here. Refer to the overall architecture diagram shown below :



Azure Logic Apps will be utilized for orchestrating the flow and automate the transformation of data between the layers of the Azure Data Lake Store. They are available in Resource group - aws-npe-prd-awbi-middleware-rg-n-eu

There are multiple Logic Apps created to orchestrate the entire flow for Telematics and Data Science. The logic apps are responsible for executing the data factory pipelines which in turn will transform data from one layer to the other .

Planned Orchestration

Currently, there is no fixed strict dependency management built into the framework. In order to automate processing, adequate time will be provided for each processing layer to be completed. There are obvious downsides to this approach, in that there is no guarantee that each layer would have completed its processing in the set amount of time. Issues from a large unexpected data load or issues with processing can occur which could be to the detriment of any downstream processing. This will require close monitoring whilst a more robust solution is put in place.

Source to Raw

The source to raw processing is intentionally decoupled from the rest of the curation process as this is to be scheduled instead of event driven, whilst also allowing the same process to work for both scheduled batch driven ingestion and replication from the SAP source.

The following diagram highlights the process used for integrating data from source systems to the [Storage Account](#)  RAW layer using [Azure Logic Apps](#) .



Name	Last Modified
 sap	Wed, 20 Mar 2019 05:34:03 GMT

Name	Access Tier	Access Tier Last M
unprocessed		

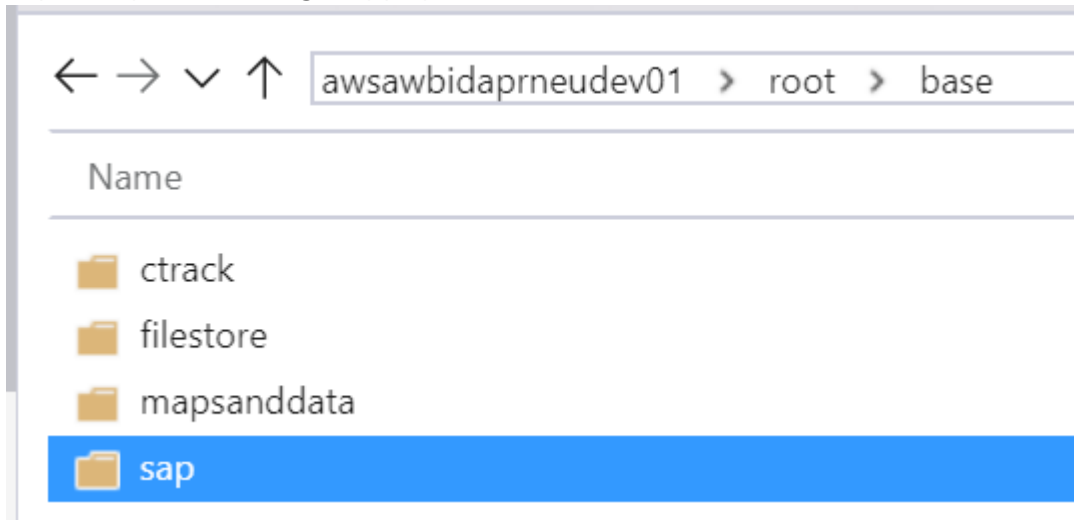
Data is available in the Raw\Unprocessed folder in Data Lake and Storage Account once the workflows defined in the Logic Apps completed successfully.

Logic App :

1. `awsawbifilestoreopfssourcetorawdev01`
2. `awsawbifilestoresftpsourcetorawdev01`

Raw to Base

Separate processes Logic Apps per source are defined which will transform and move the data from Raw to Base to the Data Lake Base Layer.



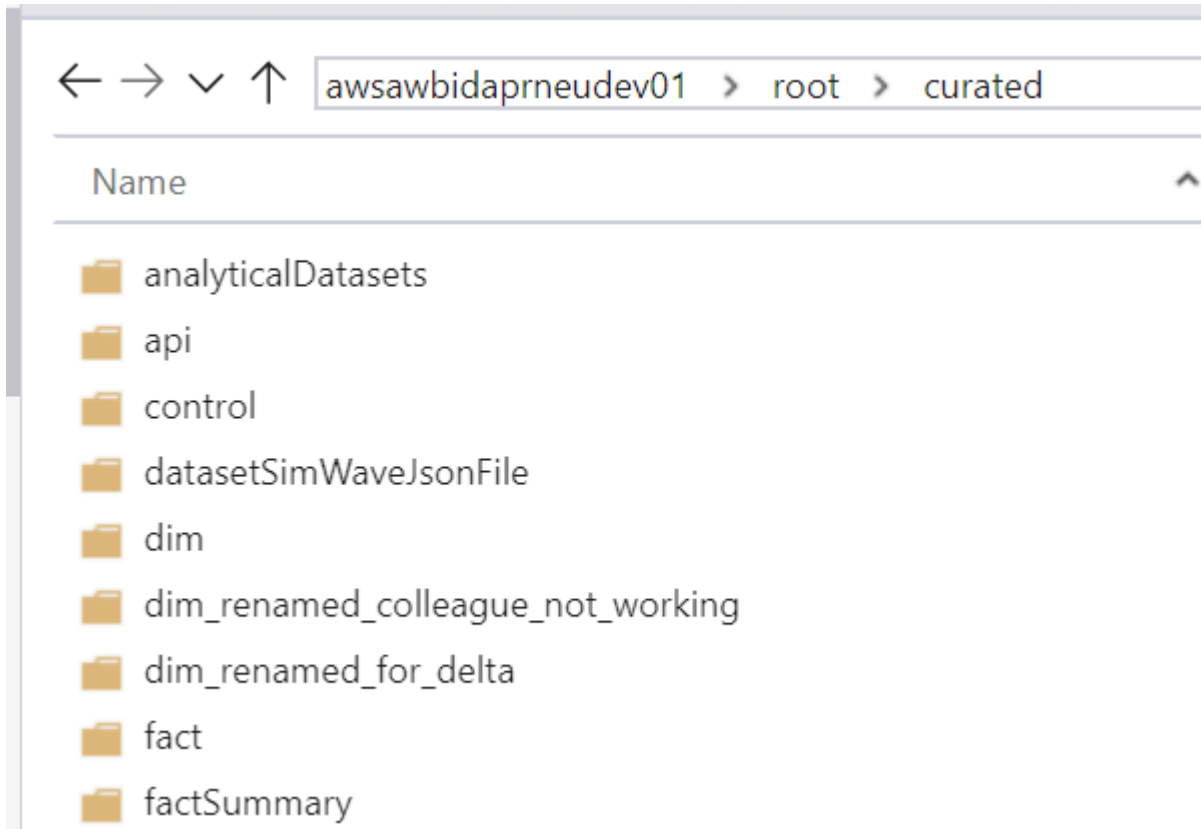
Logic Apps :

These logic apps will remain the same and new logic apps to be added in case a different source system is included in the platform in future.

1. `awsawbiattunityrawtobaseprd01`
2. `awsawbictrackrawtobaseprd01`
3. `awsawbifilestoreraawtobaseprd01`

Base to Curated

This layer will be generic and new apps needs to be included in the platform as per the epic requirement.Below logic apps are created for Telematics and Data Science.



Logic Apps :

- Telematics

1. `awsawbiprocessdimensionsprd01` - Processes all the Dimensions for Telematics
2. `awsawbiprocessfactsprd01` - Processes all the Facts for Telematics

- Data Science

As and when new epics related to Data Science gets created in the platform , new logic apps will be added. For Data Science SIM 2 Model the

below logic apps transforms and scores the data.

1. `awsawbisimwave2transformationprd01` - Transforms and creates the dataset which needs to be scored
2. `awsawbisimwave2scoreprd01` - Executed the machine learning model and copies the scored dataset to the Storage Account

Curated to Semantic

This is generic again, when new semantic models gets included in the platform, either the code needs to be updated to incorporate all the scenarios or new logic app per semantic model needs to be created. Example for telematics semantic model is mentioned below :

`awsawbiprocesssemanticprd01` - Processes the semantic model for Telematics

Overall Flow and logic app details :

